Smart AI Powered Vision Glasses: Enhancing Accessibility for the Visually Impaired

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Abstract: This project aims to develop Smart AI Powered Vision Glasses to assist visually impaired individuals in understanding their surroundings. The objective is to integrate computer vision and embedded systems for real-time object detection and obstacle awareness. The system uses a camera, sensors, and audio feedback to provide information about nearby objects and obstacles. It is designed as a portable, user-friendly, and offline wearable device to enhance safety, independence, and ease of navigation in real-world environments.

Individual contribution and findings:

I, Utkarsh Raj, was responsible for the programming and computer vision aspects of the project. My work involved implementing the circuit and the YOLOv4-Tiny object detection model using OpenCV, integrating the Raspberry Pi Camera Module, and developing the real-time detection pipeline. I also worked on system configuration, hardware implementation, debugging camera-related issues, and implementing audio feedback using text-to-speech.

Individual contribution to project report preparation:

I, Utkarsh Raj, contributed to the preparation of the project report by working on **Chapter 3: Experimentation and Tests**. I also contributed to the **Appendix (Code section)**, verified the report content, and assisted in the final compilation and submission.

Individual contribution for project presentation and demonstration:

I, Utkarsh Raj, contributed to preparing the presentation by explaining the working of the project and was actively involved in demonstrating the hardware and system functionality.

Full Signature of Supervisor/s: …….

Full signature of the student: ……..